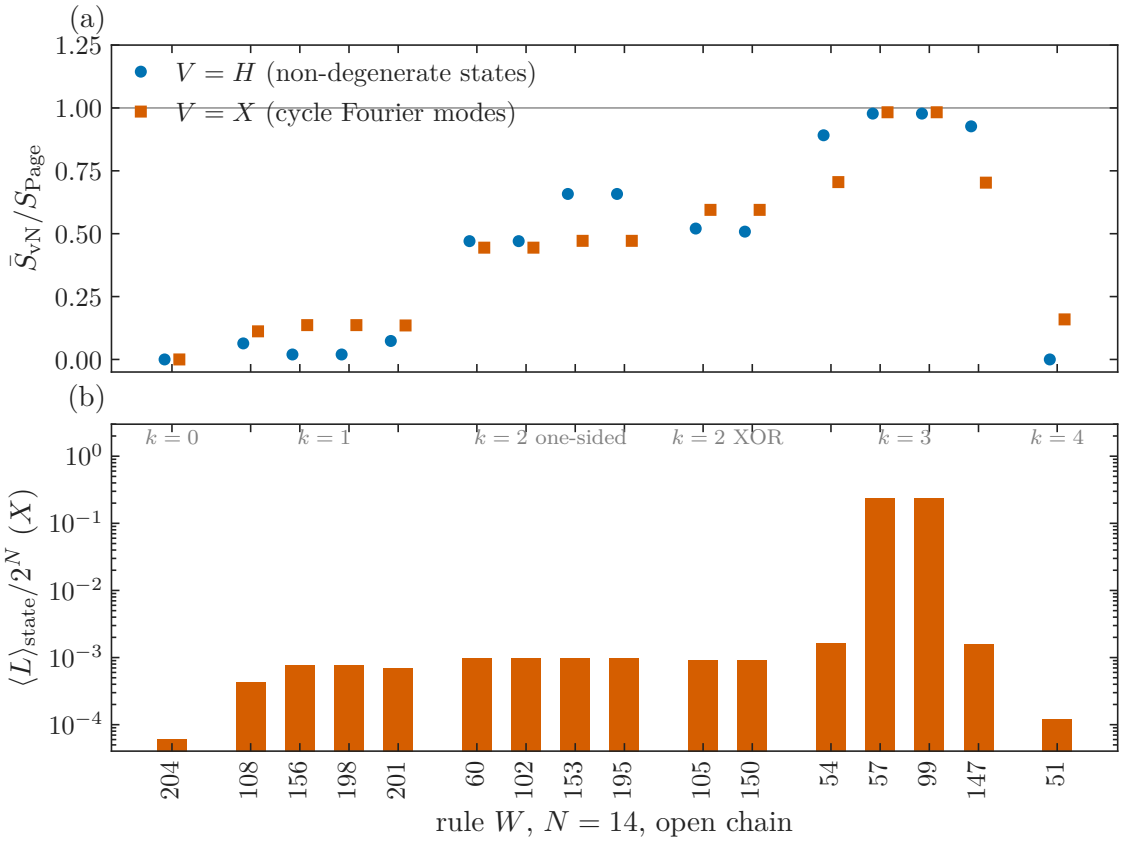

(a)
1.25
1.00
0.75
0.50
0.25
0.00
$V = H$ (non-degenerate states)
$V = X$ (cycle Fourier modes)
$\bar{S}_{\mathrm{vN}}/S_{\mathrm{Page}}$
(b)
$k = 0$
$k = 1$
$k = 2$ one-sided
$k = 2$ XOR
$k = 3$
$k = 4$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$\langle L \rangle_{\mathrm{state}}/2^{N}$ (X)
204
108
156
198
201
60
102
153
195
105
150
54
57
99
147
51
rule W, $N = 14$, open chain